Saun-Jay Samuels

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EDUCATION

Florida Gulf Coast University, Fort Myers, FL — Expected May 2027

B.S. Bioengineering (ABET-accredited), Minor in Spanish | GPA 3.50

President, Kappa Sigma Rho Zeta, Nov 2025 – Nov 2026; Collegiate 100 member; NSBE and BMES member

EXPERIENCE

Mechanical Engineer Intern, AXI International, Fort Myers, FL — May 2026 – Present

- Standardized a 60-configuration fuel transfer system product line, with assemblies and 30 customer drawings
- Restructured drawings into modular subassembly drawings combined into build packages that define assembly order
- Built a parametric Excel Bill of Materials (BOM) rollup to price 60 configurations (389 components, 11 subassemblies)
- Drove 100+ drawings through peer and final review, resolving redlines and auditing BOMs against drawings
- Inspected 8 field-return systems with torque and dimensional checks and documented rework for engineering review
- Tested 4 controllers and wired them to test 40 solenoid-operated valves
- Prepared UL 142 day tank fabrication drawings for 6 tank sizes with UL traceability documentation
- Completed a PLC fabrication feasibility study and standardized SolidWorks PDM

Power Delivery: Distribution Intern, Florida Power & Light (NextEra Energy), Naples, FL — May 2025 – Aug 2025

- Completed 216 site assessments, using GIS and parcel data to classify service requests
- Produced 25+ overhead and underground designs with one-line diagrams and job packages
- Calculated transformer loading and voltage drop and completed 10 avian protection retrofits

Undergraduate Researcher, Biomaterials Lab (Dr. Jiehong Liao), FGCU — Aug 2025 – May 2026

- Fabricated PCL nanofiber scaffolds by solution blow spinning and optimized process parameters
- Characterized scaffolds with Instron tensile testing and SEM, and quantified fiber diameter in MATLAB and ImageJ
- Assessed cell adhesion, proliferation, and viability on the scaffolds

Facility Manager, FGCU Recreation and Wellness — Apr 2024 – Present

- Supervise and train 30+ staff and enforce safety protocols

PROJECTS

Inclusive Pulse Oximeter, Senior Design (Team Lead of 4) — Aug 2026 – May 2027 (expected)

- Proposed a project to reduce SpO2 measurement bias across skin tones and lead a four-person team, owning research and system integration under a Design History File (DHF) framework

Autonomous Ball-Sorting Robotic Arm — Mar 2026 – May 2026

- Programmed a 3-DOF arm with servo gripper in C++ on Arduino, using a 150 degree ultrasonic scan and RGB color classifier to sort objects, and designed the 3D-printed detection system in SolidWorks

Slide-Track Bag Opener (Assistive Packaging) — Apr 2026

- Converted 9 assisted-living user interviews into target specifications and a DHF with risk analysis, then designed a support-free FDM part under 30 g

Aluminum Cube Assembly (Team of 6)

- Machined parts on CNC equipment, designed the fixtures to hold them, and inspected finished parts with LiDAR and calipers

Adjustable Laptop Cooling Stand

- Machined a 6061-T6 stand to ±0.13 mm with an FEA factor of safety of 9.7 and GD&T drawings

SKILLS

CAD and Design: SolidWorks (CSWE, CSWP; PDM, Simulation, sheet metal, weldments), Fusion 360, Inventor, AutoCAD, Geometric Dimensioning and Tolerancing (GD&T), DFM, Finite Element Analysis (FEA)

Manufacturing and Test: CNC milling, fixture design, FDM 3D printing, Instron, SEM, oscilloscope, strain gauges, Wheatstone bridges, ISO 10993-5, ANOVA

Programming and Quality: Python, C++, Arduino, MATLAB, Simulink, R, ImageJ; design controls, Design History File, configuration management, Lean Six Sigma White Belt